

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY

(Deemed to be University), Girinagar, Pune - 411025

M. Tech/PhD EXAMINATIONS

THIRD MID SEMESTER EXAMINATION - (July - Dec) - 2022

Course (Subject): Advanced Numerical Methods

Course Code: AM 601

Maximum Marks: 10

Duration: 01 hrs

1. Solve $u_{tt} = 4 u_{xx}$ with boundary conditions $u(0, t) = 0 = u(4, t)$, $t > 0$ and the initial conditions $u_t(x, 0) = 0$, $u(x, 0) = x(4 - x)$

[4M]

2. By using the Lax-Wendroff Implicit Formula, solve the equation $5 u_x + 6 u_y = 10$, $0 < x < \infty$, $y > 0$ subject to the conditions $u(x, 0) = 0$, $0 \leq x \leq 5$, $u(x, 0) = \sin \pi x$, $x \geq 5$, $u(0, y) = 0$, $0 < y < \infty$, with, $h = 1/3$, $k = 1/3$.

[3M]

3. Find the largest Eigen value and its corresponding Eigen vector of the flowing matrix by using Power method.

$$\begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & 1 \\ 1 & 1 & 3 \end{pmatrix}, \text{ by taking } X_0 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$C=4$

$\lambda = \frac{k^2 c^2}{h^2}$

$\frac{16}{(\frac{1}{4})^2} = \frac{1}{4}$

$\lambda = \frac{k^2 c^2}{h^2}$

$\times 16 \times 16$

[3M]

$\frac{3}{4} + \frac{2}{3} + \frac{2}{7} =$

a_1

$\frac{2}{3} =$

$\frac{2.66}{3}$

$\frac{2.66}{3}$

$\frac{1.7}{16}$

$\frac{1.7}{16}$

$\frac{1.7}{16}$

$\frac{1.7}{16}$

$\frac{1.7}{16}$

$\frac{1.7}{16}$

$\frac{2 \times 3}{4} + \frac{2}{3} + 1$

$\frac{6.667}{2.667}$

$\frac{1.9}{6}$

$\frac{3}{4} + \frac{2}{3} + \frac{3}{7}$

$\frac{4.0}{5.12}$

$\frac{2}{3} + \frac{1}{3} + \frac{3}{7}$

$\frac{8}{3} =$

$\frac{2.66}{3}$

$\frac{2.3+9}{3}$

$\frac{4}{3}$

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY
(Deemed to be University), Girinagar, Pune - 411025
M.Tech/PhD EXAMINATIONS
FIRST MID SEMESTER EXAMINATION-(AUTUMN:2022)

Course(Subject): **Advanced Numerical Methods**

Course Code: **AM 601**

Maximum Marks: **10**

Duration : **01 hrs**

- Please ensure that you have the correct question paper.
- Marks are indicated against each question.
- Write clearly and legibly.

1. Why is the classical Runge-Kutta method of fourth order, the most commonly used method for solving the first order initial value problems? (1M)
2. What is the disadvantage of the Taylor's Series method. (1M)
3. Solve by *Euler's* method the differential equation $\frac{dy}{dx} = \frac{y-x}{y+x}$ at $x = 0.1$ correct to four decimal places with the initial condition $y(0) = 1$ with step size 0.02 (3M)
4. Obtain the solution of the Initial Value Problem $\frac{dy}{dx} = x^3 + y$, $y(0) = 2$, then compute $y(0.2)$, $y(0.4)$ and $y(0.6)$ by using R-K method of order 4. Compare the result with exact solution. (5M)

1st ATE

$$K_1 = hf(x_0, y_0)$$

$$K_2 = hf\left[x_0 + \frac{h}{2}, y_0 + \frac{K_1}{2}\right]$$

$$K_3 = hf\left[x_0 + \frac{h}{2}, y_0 + \frac{K_2}{2}\right]$$

$$K_4 = hf[x_0 + h, y_0 + K_3]$$

Sum

$$y_1 = y_0 + \frac{K_1 + K_2 + K_3 + K_4}{4}$$

4th order

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY

(Deemed to be University), Girinagar, Pune - 411025

M. Tech/PhD EXAMINATIONS

SECOND MID SEMESTER EXAMINATION - (July - Dec) - 2022

Course (Subject): Advanced Numerical Methods

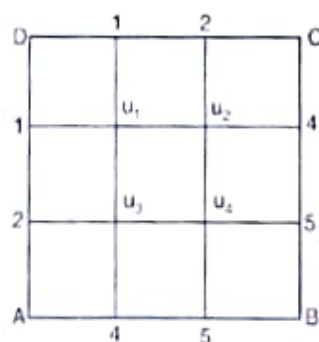
Course Code: AM 601

Maximum Marks: 10

Duration: 01 hrs

Note:

- Before attempting to answer please ensure that you have the correct question paper.
 - Calculators are only allowed.
 - Write clearly and legibly. All diagrams are neatly to be labelled wherever applicable.
 - Answer all questions. Marks are indicated against each question.
-
- What is the condition of stability for the Schmidt method? Is the Bender-Schmidt method for solving the heat conduction equation stable? What type of system of equations do we get when we apply the Crank-Nicolson method to solve the one-dimensional heat conduction equation?
[1M]
 - Obtain the solution of the Initial Value Problem $\frac{dy}{dx} = x^2 + x^2y$, $y(1) = 1$ at $x = 1$ (0.1) 1.3 by any numerical method you know, and at $x = 1.4$ by Adams Bashforth P-C method. Compare the values.
[3M]
 - Using the Liebmann method, solve the equation $U_{xx} + U_{yy} = 0$ for the following square mesh with boundary values as shown in figure. Iterate until the maximum difference between successive values at any point is less than 0.001.
[4M]



- Solve $u_{xx} = u_t$ subject to the conditions $u(x, 0) = \sin \pi x$, $0 \leq x \leq 1$, $u(0, t) = u(1, t) = 0$ using the Crank-Nicolson method with, $h = 1/3$, $k = 1/36$. Do one time step.
[2M]

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY

(Deemed to be University), GIRINAGAR, PUNE - 411025

M.Tech. / PhD END SEMESTER EXAMINATION: AUTUMN 2022

Course Name: Advanced Numerical Methods

Course Code: AM 601

Duration: 03 Hrs

Max Marks: 50

Note:

1. Before attempting please ensure that you have the correct question paper.
2. Marks are indicated against each question. Calculators are only allowed.
3. Write clearly and legibly. All diagrams to be neatly labeled wherever applicable.
4. Answer all questions.

Q1)

[5 + 5 = 10M]

- a) When do we use the power method? When can we expect faster convergence in power method? Does the power method give the sign of the largest Eigen value? Determine the numerically largest Eigen value and the corresponding Eigen vector of the following matrix, using the power method upto four decimal place accuracy

$$\begin{pmatrix} 25 & 1 & 2 \\ 1 & 3 & 0 \\ 2 & 0 & -4 \end{pmatrix}$$

- b) Approximate the 'y' value at $x = 0.4$ of the following differential equation by Adams-Moulton method. $\frac{dy}{dx} = 0.5y, y(0) = 1.0$

[5 + 5 = 10M]

Q2)

- a) By using the Lax-Wendroff Implicit Formula, solve the equation $u_x + u_y = 6, 0 < x < \infty, y > 0$ subject to the conditions $u(x, 0) = x, 0 \leq x \leq 2, u(x, 0) = \sin \pi x, x \geq 2, u(0, y) = 0, 0 < y < \infty$, with $h = 1, k = \frac{1}{2}$.
- b) Using Milne's predictor-corrector method, find $y(0.4)$ for the initial value problem $y' = x^2 + y^2, y(0) = 1$, with $h = 0.1$. Calculate all the required initial values by Euler's method. The result is to be accurate to three decimal places.

[4 + 6 = 10M]

Q3)

- a) Given $y = x^3 + y, y(0) = 2$. Compute $y(0.2), y(0.4)$ and $y(0.6)$ using the Runge-Kutta method of fourth order up to five decimal places.
- b) Solve $u_{tt} = 4u_{xx}$ with boundary conditions $u(0, t) = 0 = u(4, t), t > 0$ and the initial conditions $u_t(x, 0) = 0, u(x, 0) = x(4 - x)$

Q4)

[10M]

Write the Implicit method for solving the one dimensional heat conduction equation. What is the order and truncation error of the method? What is the condition of stability for this method. Hence solve the below mentioned 1D heat conduction equation subject conditions

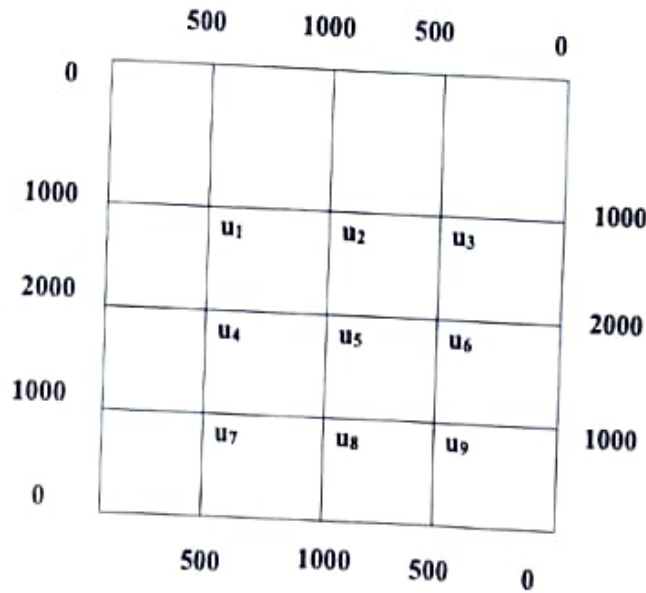
$$u_t = u_{xx} \quad 0 \leq x \leq 1, \text{ with } u(x, 0) = \sin 2\pi x \leq x \leq 1, u(0, t) = u(1, t) = 0$$

Assume $h = 0.25$, $\lambda = 0.8$ Compute for two time steps. Integrate for two time steps. If the exact solution of the problem is $u(x, t) = e^{-4t\pi^2} \sin 2\pi x$. Find the magnitudes of the errors on the second time step.

Q5)

[10M]

Using the Liebmann method, solve the equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ for the following square mesh with boundary values as shown in figure. Iterate until the maximum difference between successive values at any point is less than 0.001.



DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY

(Deemed to be University)

M.Tech/MS/PhD Examinations

Assessment-I-Examination (Autumn 2022)

Course Name: Mathematical Modelling and System Analysis

Course Code: AM 602

Duration: 01 hr

Date:27/09/2022

Max Marks:10

Note:

1. Marks are indicated against each question. Write clearly and legibly.
All diagrams to be neatly labeled. Attempt all questions. Calculators are allowed.
 2. Before attempting please ensure you have the correct question paper.
-

Q1. Consider a mathematical model suitable to describe the competition between predators P and prey p homogeneously distributed in a certain territory. In particular, suppose that the model is the following

$$\begin{cases} \frac{dP}{dt} = c_1 P + c_2 p \\ \frac{dp}{dt} = c_3 p + c_4 P. \end{cases}$$

- i. Define the dependent and the independent variables of such a model.
- ii. Define the parameters.
- iii. Is the model a discrete or a continuous one?
- iv. Is the model static or dynamic one?
- v. Are the constants c_i , $i = 1, \dots, 4$, which characterize the model, positive quantities? Why?
- vi. Assume that the following terms

$$c_5 P^2, \quad \text{and} \quad c_5 < 0.$$

- is added to the first equation. What is the physical meaning of such a term?
- vii. Assume that the term $g(t) \geq 0$ is added to the second equation. What is the physical meaning of such a term?
 - viii. Consider the mathematical model for the dynamics of predators P and prey p nonhomogeneously distributed in a two-dimensional territory.
Write the mathematical model for the above corresponding equations.
-

ALL THE BEST

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY

(Deemed University)

M.Tech/MS/PhD Examinations

End Sem Examination (Autumn 2022)

Course Name: **Mathematical Modelling and System Analysis**

Course Code: **AM 602**

Duration: 03 hrs

Max Marks:50

Note:

1. Marks are indicated against each question. *Write clearly and legibly.* All diagrams to be neatly labeled. Attempt all questions. Calculators are allowed.
2. Before attempting please ensure you have the correct question paper.

Question:1 (A) Consider the one-dimensional convection-diffusion model

$$\frac{\partial u}{\partial t} + c(u) \frac{\partial u}{\partial x} = \frac{\partial}{\partial x} \left(h(u) \frac{\partial u}{\partial x} \right) + g(t, x, u).$$

- i) Explain why the model is a dynamic continuous one.
- ii) Explain why the model is nonlinear.
- iii) What are the conditions required to characterize such a model as a static one?
- iv) What are the conditions required to characterize such a model as a discrete one?
- v) What are the conditions required to characterize such a model as a linear one?

(B) Consider the initial value problem

(5 marks)

$$\frac{du}{dt} = A(t)u + b(t) \quad \text{with} \quad u(t_0) = u_0,$$

where $A_{ij}(t)$, and $b_j(t)$ are continuous for $|t - t_0| \leq T$. Do this problem have unique continuous solution? If yes, find the interval where it has a unique solution. Also, find the Lipschitz constant.

(C) Find the analytical solution for the following competition model

(3 marks)

$$\frac{du}{dt} = \alpha u - \beta v \quad \frac{dv}{dt} = \gamma v^2,$$

with $u_0 = 1$, and $v_0 = 1$. Where α , and β are positive constants and γ may take both positive and negative values.

(2 marks)

Question:2 Consider the initial value problem (IVP) $\frac{du}{dt} = f(t, u)$, with $u(t_0) = u_0$

(A) Solve the IVP by backward Euler-method, find the truncation error, and write the MATLAB (or any computer language) algorithm/code/pseudocode for IVP. (5 marks)

(B) Suppose $f(t, u)$ is nonlinear, then which method will you prefer to solve the IVP among the forward Euler or backward Euler method? Why? (3 marks)

(C) Discuss the idea of Adams-Bashforth or Adams-Moulton method (any one) to solve the IVP. Solve the IVP by any of the Adams-Bashforth or Adams-Moulton method of order two. (2 marks)

Question:3 (A) Write down the numerical advantages and disadvantages of Parallelisation in parallel computing. Consider the problem $\frac{d^2u}{dx^2} + u = f(x)$, $x \in (0, 2)$, with boundary conditions $u(0) = 1$ and $u(2) = 1$. Solving the problem using two machines with proper justifications. (5 marks)

(B) What are the explicit and implicit finite difference schemes to solve the initial value problems or boundary value problems. Give examples of each method. And, write the advantages and disadvantages of explicit and implicit methods over each other. (3 marks)

(C) Using the predictor and corrector method solve $\frac{du}{dt} = f(t, u)$, with $u(0) = u_0$. (2 marks)

Question:4 (A) Consider the one dimensional scalar transport equation

$$\frac{\partial u}{\partial t} + A(t, x) \frac{\partial u}{\partial x} = f(t, x; u), \quad x \in [0, 1],$$

where the eigenvalues λ_i of $A(t, x)$ all have same sign with the initial condition $u(0, x) = u_I(x)$, and the boundary condition $u(t, 0) = u_0(t)$. (Classify and develop the Leap-Frog finite difference method to solve the given model. Also find the stability region for the method.) (5 marks)

(B) The dynamical behaviour governing the growth, decay and general evolution of two interacting biological species is modelled by the Lotka-Volterra prey-predator equations, which are given by

$$\begin{cases} \frac{dx}{dt} = ax - bxy \\ \frac{dy}{dt} = -cy + dxy. \end{cases}$$

Here x denotes the population of the prey, say rabbits at a given time and y denotes the population of the predator, say foxes. The constants a, b, c, d are all positive and represent the growth and decay rates of the prey and the predator. Discuss existence and uniqueness of the given model. Also modify the model to incorporate the completion within the same species with dynamic changes. Justify all the parameters in the modify model. (5 marks)

Question:5 (A) Consider 1D motion of an object falling in air under the Newton Drag Law $m \frac{dv}{dt} = -mg - bv^2$, (v assumed > 0). Then find the dimension and write the non-dimensional form. (4 marks)

(ii) Write the significance of the Buckingham's π theorem. (2 marks)

(B) Discuss the importance of system analysis and design in the development of a system. (4 marks)

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY

(Deemed to be University)

M.Tech /Ph.D END SEMESTER EXAMINATIONS : Nov. / Dec. 2022

Course (Subject) : **Advanced Optimization Techniques** / Course Code **AM-603**

Duration : **03 hrs**

Max Marks : **50**

Note: 1 Before attempting , please ensure that you have the correct question paper.

2. Attempt **all Five** questions

3. Students may request the invigilator for required Tables (if any).

1). Solve the following L.P.P. by Simplex method (10 M)

Maximize $Z = -x_1 + 3x_2 - 2x_3$

S.to. $3x_1 - x_2 + 2x_3 \leq 7$

$-2x_1 + 4x_2 \leq 12$

$-4x_1 + 3x_2 + 8x_3 \leq 10$

$x_1, x_2, x_3 \geq 0$

2). Using Kuhn-Tucker conditions to solve the following NLPP. (10 M)

Max $Z = -x_1^2 - x_2^2 - x_3^2 + 4x_1 + 6x_2$

S.to. $x_1 + x_2 \leq 2$

$2x_1 + 3x_2 \leq 12$

$x_1, x_2 \geq 0$

3. a). Find the extreme points and their nature of the function (5+5 = 10 M)

$f(x_1, x_2) = x_1^3 + x_2^3 + 2x_1^2 + 4x_2^2 + 6$

b). Write a short note on Genetic Algorithm and its applications.

4.. Use Steepest Decent Method to solve the following problem. (10 M)

Minimize $f(x_1, x_2) = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2$ starting from the point $X_1 = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$

5. A small project is composed of seven activities whose time estimates are listed in the table as follows: (10 M)

Activity		1-2	1-3	1-4	2-5	3-5	4-6	5-6
time estimates in weeks	t_o	5	1	2	3	1	2	1
	t_m	6	1	4	6	1	2	4
	t_p	7	7	12	15	1	8	7

a). Draw the Project net work

b). Find the critical path of the project and project duration.

c). What is the probability that the project will be completed at least 4 weeks earlier than expected. duration.

----- @@@@ -----

9

1 > 4

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY
(Deemed to be University)

M.Tech/ Ph.D (Autumn Semester End Exam Dec. 2022)

Subject : Statistical Computing for Data Science

Course Code : AM 604D

Duration : 03 Hrs.

Max. Marks : 50

- a. Marks are indicated against each question.
- b. Write clearly and legibly. All diagrams to be neatly labelled.
- c. **Calculators and Statistical tables are allowed.**
- d. Before attempting please ensure that you have the correct question paper

1. Let X_1, X_2 be two random variables with covariance matrix $\begin{bmatrix} 9 & \sqrt{6} \\ \sqrt{6} & 4 \end{bmatrix}$. Obtain the principal components and find the percentage of total error explained by each component. [10]
2. Given the following set of data find out the multiple regression plane and check whether the regression is significant at 0.05 significance level. [10]

Y	10.98	7.82	8.11	10.36
X_1	35.3	70.7	70	33.4
X_2	20	21	22	20

3. a) Discuss in detail about Linear and Non-Linear regression models with examples. [6]
- b) Explain about Logistic regression and Poisson Regression. [4]
4. a) State central Limit theorem. [2]
- b) Find the probability of getting more than 25 sevens in 100 tosses of a pair of fair dice. [3]
- c) Find the 99% confidence intervals for the variance and variance ratios of normally distributed populations. [5]
5. The mean height of 50 male students who showed above average participation in college athletics was 68.2 inches with a standard deviation of 2.5 inches, while 50 male students who showed no interest in such participation had a mean height of 67.5 inches with a standard deviation of 2.8 inches. Test the hypothesis that male students who participate in college athletics are taller than other male students. Also find P value of the test. [10]

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY

(Deemed to be University)

M.Tech/ Ph.D (Autumn Semester End Exam Dec. 2022)

Subject : Advanced Statistical Techniques

Course Code : AM 604

Duration : 03 Hrs.

Max. Marks : 50

- a. Marks are indicated against each question.
- b. Write clearly and legibly. All diagrams to be neatly labelled.
- c. **Calculators and Statistical tables are allowed.**
- d. Before attempting please ensure that you have the correct question paper

$$S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y})$$

$$S_{xx} = \sum (x_i - \bar{x})^2$$

$$S_{yy} = \sum (y_i - \bar{y})^2$$

1. a) Discuss briefly about Random Processes. .
- b) Explain Stationary process and Markov chain with examples. .
- c) Discuss in detail about Linear and Non-Linear regression models with examples. [10]
- ✓ 2. Given the following set of data find out the multiple regression plane and check whether the regression is significant at 0.05 significance level. [10]

286

Y	10.98	7.82	8.11	10.36
X ₁	35.3	70.7	70	33.4
X ₂	20	21	22	20

3. a) State central Limit theorem. [2]
- ✓ b) Find the probability of getting more than 25 sevens in 100 tosses of a pair of fair dice. (146) [3]
- c) Find the 99% confidence intervals for the variance and variance ratios of normally distributed populations [5]
4. The mean height of 50 male students who showed above average participation in college athletics was 68.2 inches with a standard deviation of 2.5 inches, while 50 male students who showed no interest in such participation had a mean height of 67.5 inches with a standard deviation of 2.8 inches. Test the hypothesis that male students who participate in college athletics are taller than other male students. Also find P value of the test. (1.71) 228 [10]

- ✓ 5. a) Derive the optimal values of the parameters in case of simple regression. [6]
- b) Complete the missing terms in the following ANOVA Table of simple regression [4]

$SSR = \frac{S_{xy}^2}{S_{xx}}$
 $SSE = S_{yy} - SSR$

	df	SS	MS	F	Significance F
Regression	1 ?	SSR ?	msr ?	. ?	5.76708E-08
Residual	n-2 ?	9.18886087	ml ?		
Total	n-1 9	429.709			

$MSR = \frac{SSR}{1}$

$MSR = \frac{SSE}{n-2}$

$F_{ratio} = \frac{MSR}{ML}$

$\frac{1.001}{(25) / 7.51}$

$\frac{n_1}{n_2} (n-2)$

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY

(Deemed to be University)

M.Tech/MS/PhD Examinations

Assessment-I-Examination (Autumn 2022)

Course Name: Mathematical Methods

Course Code: AM 606

Duration: 01 hr

Date: 30/09/2022

Max Marks: 10

Note:

1. Marks are indicated against each question. Write clearly and legibly.
All diagrams to be neatly labeled. Attempt all questions. Calculators are allowed.
2. Before attempting please ensure you have the correct question paper.

✓ Q1. Let $y'' + p(x)y' + q(x)y = 0$. Suppose $y_1(x)$ and $y_2(x)$ are two solutions of given differential equation. Show that linear combination of two solutions $(y_1(x), y_2(x))$ on an open interval I , is again a solution of given differential equation on I .

✓ Q2. Solve

$$y \frac{dy}{dx} + (x+1)y = e^{x^2} y^3$$

$$u = \frac{1}{y^2} \quad \frac{du}{dx} = -\frac{2}{y^3} \frac{dy}{dx}$$

✓ Q3. Solve

Let $(\sin(y)\cos(y) + x \cos^2(y)) dx + x dy = 0$. Test for exactness. If it is not exact then find an integrating factor and solve it.

✓ Q4. Check the existence and uniqueness of the solution for the initial value problem

$$\frac{dy}{dx} = y^2, \quad y(0) = 1.$$

Also find the solution in the appropriate interval.

ALL THE BEST

$$\frac{dy}{dx} + Py = Q$$

$$I.F. = e^{\int P dx}$$

$$I.F. y = \int (I.F.) Q dx + C$$

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY

(Deemed University)

M.Tech/MS/PhD Examinations

Examination (Autumn 2022)

Course Name: **Mathematical Methods**

Course Code: **AM 606**

Duration: 03 hr

Max Marks: 50

Note:

1. Marks are indicated against each question. Write clearly and legibly. All diagrams to be neatly labeled. Attempt all questions. Calculators are allowed.
2. Before attempting please ensure you have the correct question paper.

Question:1

[4 + 4 + 2 Marks]

- (a) Show that $f(x) = |x|^{1/2}$ is not locally Lipschitz at 0, that is, f is not Lipschitz in any interval (a, b) containing the origin. But, it is Lipschitz in any interval (finite or infinite) away from the origin. Is it Lipschitz in $(0, 1)$? Justify your answer.

- (b) Solve $\frac{d^2 y}{dx^2} - y = 0$ for the initial conditions $y(0) = 1$, $\frac{dy}{dx}(0) = -1$. Then change the initial conditions to $y(0) = 1.001$, $\frac{dy}{dx}(0) = -0.999$ and explain why this small change of 0.001 at $t = 0$ causes a larger change later.

- (c) Find the homogeneous solution for the problem $\frac{d^4 y}{dx^4} - 4\frac{d^3 y}{dx^3} + 6\frac{d^2 y}{dx^2} + y = e^x \sin(x)$.

Question:2

[4 + 3 + 3 Marks]

- (a) Consider the linear equation $\frac{d^2 y}{dx^2} - \frac{1+x}{x} \frac{dy}{dx} + \frac{1}{x} y = xe^{2x}$. By inspection, we observe that $y_1(x) = e^x$ is a solution of the homogeneous equation. Using the method of reduction order find the $y_2(x)$ for the homogeneous equation. Then find a particular solution $y_p(x)$ of the given non-homogeneous equation.

- (b) Using the method of variation of parameters to determine the general solution of the given differential equation $\frac{d^3 y}{dx^3} + \frac{dy}{dx} = 2x^2 + 4\sin(x)$.

- (c) Find a power series solution of $(1-x^2)\frac{d^2 y}{dx^2} - 2x\frac{dy}{dx} + 2y = 0$.

Question:3

[3 + 3 + 4 Marks]

- (a) Find a basis for a subspace U of V , where $U = \{p \in P_5 \mid p'' = 0\}$ and $V = P_5$.

$$(2\sqrt{a_0})y^{20} \quad D = \frac{1}{2}\sqrt{-a_0}$$

(b) Consider the sets:

$$\begin{cases} S_1 = (1, 1, 0), \\ S_2 = (1, 1, 0), (0, 1, 1), \\ S_3 = (1, 1, 0), (0, 1, 1), (1, 0, -1), \\ S_4 = (1, 1, 0), (0, 1, 1), (1, 0, -1), (1, 1, 1). \end{cases}$$

$\rightarrow \text{LI } p(S_2) = 2$
 $p(S_4) = 2$

Find the largest linearly independent subset whose span is $[S_4]$.

(c) Let $T : P_3 \rightarrow P_2$ be a linear transformation defined by $T(a_0 + a_1t + a_2t^2 + a_3t^3) = (a_0 + a_3) + (a_1 + a_2)t + (a_0 + a_1 + a_2 + a_3)t^2$. Determine the matrix for the given linear transformation T and the bases $B_1 = \{1, t, t(t+1), t(t+1)(t+2)\}$ and $B_2 = \{1, t, t(t-1)\}$.

Question:4

[2 + 4 + 4 Marks]

(a) Suppose that A is the matrix defined as

$$A = \begin{pmatrix} 2 & 1 \\ 6 & 5 \\ 2 & 4 \end{pmatrix}.$$

Explain how knowing all solutions to $Ax = b$ decides if a given vector b in the column space of A .

(b) Reduce the matrix

Can we first apply elementary operation.

$$A = \begin{pmatrix} 7 & -1 & -10 \\ -1 & 7 & 10 \\ -10 & 10 & -2 \end{pmatrix}$$

$$\begin{pmatrix} a & b & c \\ a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}$$

to the diagonal form.

(c) Using the Gram-Schmidt process to orthonormalise the linearly independent sets of vectors $\{x^2 - 1, x + 1, x - 1\}$. (Take the inner product in the space to be $\int_{-1}^1 f(t)g(t)dt$.)

Question:5

[4 + 3 + 3 Marks]

(a) Show that the matrix

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

is not diagonalizable by proving A has only one eigenvalue and the corresponding eigenspace is one dimensional. Thus, it will not be possible to obtain two linearly independent eigenvectors.

(b) Consider the equation $\frac{d^2y}{dx^2} + a_0y = 0$ in $(0, 1)$ with boundary conditions $y(0) = a_1$, $y(1) = a_2$. Find out various cases of existence and non-existence of solutions.

(c) Using Laplace Transform solve the IVP $\frac{dy}{dt} + y = t$, $y(0) = y_0$.

2

$$\frac{dy}{dt} \rightarrow \mathcal{L} \left(\frac{dy}{dt} \right) = sY(s) - y(0) = \mathcal{L}(t) = \frac{1}{s^2}$$

$t^2 +$

$$14 - 190 = -176$$

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY

(Deemed University)

M.Tech/MS/PhD Examinations

Assessment – I - (Spring 2023)

Course Name: Data Science Tools & Techniques

Course Code: AM624D

Date: 25 – 02 - 2023

Duration: 01 Hr

Max Marks: 10

Note:

1. Marks are indicated against each question. Write clearly and legibly Attempt all questions. Calculators are allowed.
2. Before attempting please ensure you have the correct question paper.

Q1	In python is there any difference between the following: $a=123456789$ and $b=1.23456789*10^{**8}$ explain with justification. What will be the output if both variables a and b are printed?	[2.5]
Q2	ECG is the acronym for the electrocardiogram, is a sequence that always starts with the numbers 1 and 2. In this sequence, every number that succeeds is the smallest (not already present in the sequence) and shares a factor (excluding 1) with its preceding number. Every number in the ECG Sequence (besides 1 and 2) has a different index from its natural index in a normal numeric sequence. See the example below to establish the ECG Sequence Index of number 3. <pre> SEQUENCE = [1, 2] 3 = no factors shared with 2 4 = shares factor 2 with number 2 SEQUENCE = [1, 2, 4] 3 = no factors shared with 4 5 = no factors shared with 4 6 = shares factor 2 with number 4 SEQUENCE = [1, 2, 4, 6] 3 = shares factor 3 with number 6 SEQUENCE = [1, 2, 4, 6, 3] Number 3 is at Index 4 in the ECG Sequence. Given an integer n implement a function that returns its ECG Sequence Index. Examples: ecg_seq_index(3) → 4 </pre>	[2.5]

199
-42
137

	<code>ecq_seq_index(5) → 9</code> <code>ecq_seq_index(7) → 13</code>	
Q3	What is the output of the following code? (One-line answer) <pre>def mystery(m): if m==[]: return(m) else: return(mystery(m[1:]) + m[1]) mystery ([22,14,19,65,82,55])</pre>	[2.5]
Q4	WAP in python to print the factorial of a number.	[2.5]

ALL THE BEST

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY
(Deemed University)
MTech EXAMINATIONS 2022

Course: Research Methodology

Code: PGC 601

Duration: 03 hr

Max Mark: 50

Note:

1. Marks are indicated against each question.
2. Write clearly and legibly. All diagrams (if any) to be neatly labeled.
3. Make necessary assumptions wherever required.

All questions are compulsory

Q1. a) What is research and why it is important? (3 Marks)

b) What are the characteristics of Research? (3 Marks)

c) What are the different types of research and explain problem oriented research with an example from your domain? (4 Marks)

Q2. a) What is hypothesis and explain the various types of hypothesis? (3 Marks)

b) Write short notes on Ethical committee and explain the role of this committee in an organization? (3 Marks)

c) What is TRIPS agreement and what are the eight sections of TRIPS? (4 Marks)

Q3. a) Differentiate between patent and design act? (4 Marks)

b) What are the steps for protection invention? (2 Marks)

c) Write short notes on copyrights and how to prevent the infringements of copyright of electronic design, coding and figures? (4 Marks)

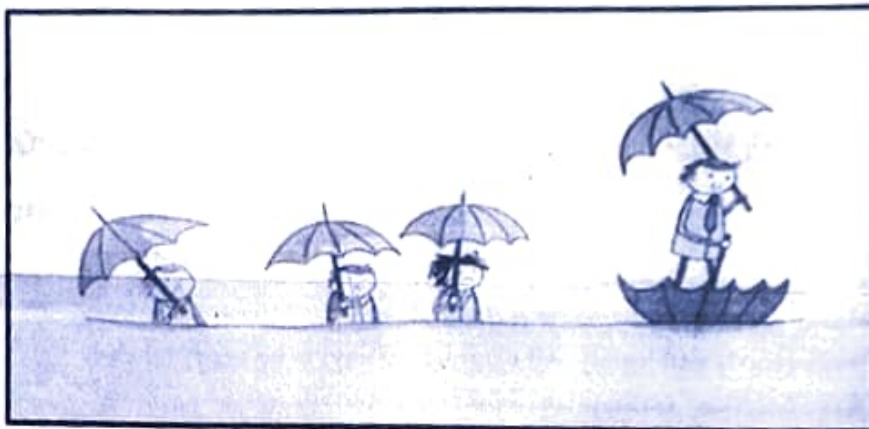
Q4.a. What are the various phases of development and explain how research methodology is important for phase development? (3 Marks)

b. What is CE certification and the various steps followed for transfer of technology? How to protect intellectual property rights? (4 Marks)

c. What is publication and the steps? What are the various types of publication? (3 Marks)

Q5. a. Comment on the schematics shown below:

(4 Marks)



Evolution

Q5. b Comment on journal article shown below

(6 Marks)

Energy
Volume 263, Part D, 15.2023, 125990

✓ **Theoretical and technical analysis of the photo-thermal energy cascade conversion for fuel with high-temperature combustion**

panelShiquanShan^a JialuTian^a BinghongChen^b YanweiZhang^a ZhijunZhou^{a*} Author links open overlay

<https://doi.org/10.1016/j.energy.2022.125990> Get rights and content

1. Who is the corresponding author and what is his role?
2. What is the title of this research and how to increase the citation index?
3. Will this article have H index?
4. What is the name of the journal and the role of first author?
5. Explain how to protect the copyright of this article?
6. Is this research patented?